ML: MULTIPLE CHOICE QUESTIONS - 1

1. In supervised learning, the training data contains:
    A) Only input data
    B) Only output data
    C) Input data with corresponding labels
    D) Random values

2. Which of the following is an example of supervised learning?
    A) Clustering
    B) Classification
    C) Dimensionality reduction
    D) Association rule mining

3. Classification is used when the output variable is:
    A) Continuous
    B) Discrete/Categorical
    C) Random
    D) Missing

4. Regression is used when the output variable is:
    A) Binary
    B) Categorical
    C) Continuous
    D) Nominal

5. Which algorithm is commonly used for classification?
    A) K-Means
    B) Decision Tree
    C) PCA
    D) Apriori

6. Which of the following is a binary classification problem?
    A) Predicting house price
    B) Detecting spam emails
    C) Forecasting sales
    D) Predicting temperature

7. In multi-class classification, the target variable has:
    A) Two classes only
    B) More than two classes
    C) No classes
    D) Infinite classes

8. Which metric is commonly used for classification evaluation?
    A) Accuracy
    B) RMSE
    C) MAE
    D) MSE

9. Accuracy is defined as:
    A) Correct predictions / Total predictions
    B) Incorrect predictions / Total predictions
    C) TP / FP
    D) FN / TP

10. Which of the following is NOT a classification algorithm?
    A) Logistic Regression
    B) Naïve Bayes
    C) KNN
    D) Linear Regression

11. KNN stands for:
    A) K-Nearest Neighbors
    B) Kernel Neural Network
    C) Known Node Network
    D) Key Number Node

12. In KNN, classification is based on:
    A) Random guessing
    B) Nearest data points
    C) Largest value
    D) Mean output

13. Decision Tree splits data based on:
    A) Rules/conditions
    B) Random values
    C) Labels only
    D) Noise

14. Logistic Regression is mainly used for:
    A) Classification
    B) Clustering
    C) Regression only
    D) Sorting

15. Which algorithm is probabilistic?
    A) SVM
    B) Naïve Bayes
    C) KNN
    D) Decision Tree

16. Which of the following is an application of classification?
    A) Face recognition
    B) Data compression
    C) Noise reduction
    D) Sorting files

17. Overfitting occurs when a model:
    A) Performs well on training but poorly on test data
    B) Performs poorly on all data
    C) Has fewer features
    D) Uses small datasets only

18. Underfitting occurs when a model:
    A) Learns noise
    B) Cannot capture patterns properly
    C) Memorizes data
    D) Has too many layers

19. Which dataset split is used to evaluate final performance?
    A) Training set
    B) Validation set
    C) Test set
    D) Input set

20. Confusion matrix is used for:
    A) Regression
    B) Classification evaluation
    C) Clustering

D) Data storage

21. Precision measures:
    A) Correct positive predictions among predicted positives
    B) Correct negatives only
    C) Total predictions
    D) Error rate

22. Recall measures:
    A) Correct positives among actual positives
    B) Correct negatives
    C) Total outputs
    D) Accuracy only

23. Which algorithm uses hyperplane separation?
    A) SVM
    B) KNN
    C) Naïve Bayes
    D) PCA

24. A model predicting Yes/No outcomes is an example of:
    A) Regression
    B) Binary Classification
    C) Clustering
    D) Reinforcement Learning

25. Which of the following is supervised learning application?
    A) Email spam filtering
    B) Market basket analysis
    C) Customer segmentation
    D) Topic modeling